

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XVIII: THE UNITARY REMAINDER IN RANK d

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ABSTRACT. Paper XV proved that for an $\tilde{A}_2$ complex of order q the geodesic edge flow satisfies $L_E L_E^\top = qI + (q^2 - q)T^\top T$, so that its remainder is $\sqrt{q}$ times an orthogonal matrix and the parahoric zeros of the edge zeta lie on $|u| = q^{-1/2}$ whether or not the complex is Ramanujan; whether this survives in higher rank was left open. It does, with the exponent $(d-1)/2$, and the proof is one line of projective geometry. In an $\tilde{A}_d$ complex the types force the tails of the edges into a vertex y to be the *hyperplanes* of the link $\text{PG}(d, q)$ and the successors to be its *points*, and $\{x, y, z\}$ is a simplex exactly when the point z lies in the hyperplane x . Since two distinct hyperplanes of $\text{PG}(d, q)$ always meet in codimension two — for every d — inclusion–exclusion gives a count that does not depend on the pair, and

$$L_E L_E^\top = q^{d-1}I + q^{d-1}(q-1)T^\top T, \quad L_E^\top L_E = q^{d-1}I + q^{d-1}(q-1)S^\top S,$$

which is Paper XV’s identity at $d = 2$. Two unconditional consequences follow with no hypothesis on the complex: *exactly* $|E| - n$ singular values of L_E equal $q^{(d-1)/2}$, and L_E restricts to a bijective $q^{(d-1)/2}$ -isometry from $\ker S$ onto $\ker T$ (the mixed identities $TL_E = q^d S$ and $SL_E^\top = q^d T$). On any subspace of $\ker S \cap \ker T$ invariant under L_E and $L_E^\top$ the remainder is therefore $q^{(d-1)/2}$ times an orthogonal matrix and its zeros lie on $|u| = q^{-(d-1)/2}$. The typing is not a convention one may vary: run the same construction on lines against planes in $\text{PG}(3, q)$ and the count takes two values, because two lines meet or are skew. Four checks, exhaustive in $\text{PG}(d, q)$ for $d \leq 5$ and reproducing Paper XV on Y_{21}, Y_{24}, Y_{42} . What is *not* done: the invariant subspace itself for $d \geq 3$, and the worked $\tilde{A}_3$ example, for want of a thick $\tilde{A}_3$ complex.

1. THE LINK, AND WHAT THE TYPES FORCE

Let X be an $\tilde{A}_d$ complex of order q in the sense of [X, Def. 1.1] generalised to rank d : a pure d -dimensional simplicial complex with vertices typed by $\mathbb{Z}/(d+1)$, one vertex of each type in every chamber, and the link of every vertex isomorphic to the flag complex of $\text{PG}(d, q)$. Write n for the number of vertices, E for the set of type-increasing directed edges ($\text{type}(y) = \text{type}(x) + 1$ for $x \rightarrow y$), and $S, T \in \{0, 1\}^{n \times E}$ for the tail and head incidence matrices. The geodesic edge flow is

$$L_E((x \rightarrow y), (y \rightarrow z)) = 1 \iff \{x, y, z\} \text{ is not a simplex of } X,$$

the rank- d form of [XV, Def. 1.1]; for $d = 2$ a 2-simplex is a chamber and this is [XV, Def. 1.1] verbatim, including the correction recorded there.

Lemma 1.1 (What the typing forces). *Fix a vertex y and identify its link with the flag complex of $\text{PG}(d, q)$, a neighbour of type $\text{type}(y) + i$ corresponding to a subspace of vector dimension i ($1 \leq i \leq d$). Then the tails of the edges into y are exactly the hyperplanes of $\text{PG}(d, q)$, the heads of the edges out of y are exactly its points, and for such x and z the set $\{x, y, z\}$ is a simplex of X if and only if the point z lies in the hyperplane x .*

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Proof. A tail x has $\text{type}(x) = \text{type}(y) - 1 \equiv \text{type}(y) + d$, so $i = d$ and x is a subspace of vector dimension d , a hyperplane; a successor z has $\text{type}(z) = \text{type}(y) + 1$, so $i = 1$ and z is a point. Two vertices of the link span a simplex with y precisely when they are incident in the link, which for a point and a hyperplane means containment. $\square$

That one paragraph is the whole difference between rank two and rank d , and it is what makes the rank-two count survive: of all the pairings of types that one might have had to handle, the typing selects the single pairing whose intersection behaviour is uniform.

2. THE IDENTITY

Write $\theta_j = \theta_j(q) = (q^{j+1} - 1)/(q - 1)$ for the number of points of $\text{PG}(j, q)$.

Theorem 2.1 (Unitarity identity in rank d). *For every $\tilde{A}_d$ complex of order q , $d \geq 2$,*

$$L_E L_E^\top = q^{d-1} I + q^{d-1}(q-1) T^\top T, \quad L_E^\top L_E = q^{d-1} I + q^{d-1}(q-1) S^\top S.$$

Proof. $(L_E L_E^\top)(e, e')$ counts the common successors of e and e' , which is zero unless e and e' have the same head y ; so the matrix is block diagonal by head, and $T^\top T$ is the indicator of “same head”. Fix y and use Lemma 1.1: the block is $BB^\top$ with $B[x][z] = 1$ iff the point z is not in the hyperplane x . On the diagonal, the number of points off a hyperplane is $\theta_d - \theta_{d-1} = q^d$. Off the diagonal, two *distinct* hyperplanes of $\text{PG}(d, q)$ meet in a subspace of codimension two, which carries θ_{d-2} points, so by inclusion–exclusion the number of points on neither is

$$\theta_d - 2\theta_{d-1} + \theta_{d-2} = (\theta_d - \theta_{d-1}) - (\theta_{d-1} - \theta_{d-2}) = q^d - q^{d-1} = q^{d-1}(q-1),$$

independent of the pair and of d . Hence each block is $q^d I + q^{d-1}(q-1)(J - I) = q^{d-1} I + q^{d-1}(q-1)J$, which is the first identity. The second is the same computation in the dual plane: two distinct points of $\text{PG}(d, q)$ span a line, the hyperplanes containing both are the θ_{d-2} hyperplanes containing that line, and the count of hyperplanes missing both points is again $q^{d-1}(q-1)$. $\square$

Remark 2.2 (Reduction to Paper XV). At $d = 2$ the constants are $q^{d-1} = q$ and $q^{d-1}(q-1) = q^2 - q$, so Theorem 2.1 is [XV, Thm. 2.1] and its proof is [XV, Lem. 1.3] with the counting done once instead of case by case. Check (A) verifies the two identities, with these constants, on the complexes Y_{21} , Y_{24} and Y_{42} of [X, Constr. 1.4].

Proposition 2.3 (The mixed identities). $TL_E = q^d S$ and $SL_E^\top = q^d T$.

Proof. $(TL_E)(v, f)$ counts the edges e with head v and $e \rightarrow f$; since $e \rightarrow f$ forces $\text{head}(e) = \text{tail}(f)$, the entry vanishes unless $v = \text{tail}(f)$, and then it counts the hyperplanes of the link of v not containing the point $\text{head}(f)$, of which there are $\theta_d - \theta_{d-1} = q^d$. The second identity is dual. $\square$

Corollary 2.4 (Unconditional consequences). *For every $\tilde{A}_d$ complex of order q :*

- (i) *exactly $|E| - n$ singular values of L_E equal $q^{(d-1)/2}$;*
- (ii) *L_E maps $\ker S$ into $\ker T$, bijectively, and $\|L_E w\|^2 = q^{d-1} \|w\|^2$ there; symmetrically $L_E^\top$ maps $\ker T$ onto $\ker S$ as a $q^{(d-1)/2}$ -isometry.*

Proof. (i) $L_E L_E^\top - q^{d-1} I = q^{d-1}(q-1) T^\top T$ has rank $\text{rank } T = n$, every vertex being the head of some edge; so q^{d-1} is an eigenvalue of $L_E L_E^\top$ of multiplicity exactly $|E| - n$. (ii) For $w \in \ker S$, $TL_E w = q^d S w = 0$ by Proposition 2.3, and $\|L_E w\|^2 = \langle L_E^\top L_E w, w \rangle = q^{d-1} \|w\|^2$ by Theorem 2.1; both spaces have dimension $|E| - n$, so the isometry is onto. $\square$

Corollary 2.5 (The critical circle in rank d). *Let $W \subseteq \ker S \cap \ker T$ be invariant under L_E and $L_E^\top$ and put $R = L_E|_W$. Then $RR^\top = R^\top R = q^{d-1} I_W$, so $q^{-(d-1)/2} R$ is orthogonal and every root of $G(u) = \det(I - uR)$ lies on $|u| = q^{-(d-1)/2}$. No hypothesis on the complex — in particular no Ramanujan hypothesis — is used.*

Proof. Immediate from Theorem 2.1, since $T^\top T$ and $S^\top S$ vanish on W ; an orthogonal matrix has all eigenvalues of modulus one. $\square$

Remark 2.6 (The typing is not a convention). The uniformity in Theorem 2.1 is a property of the hyperplane/point pairing, not of $\text{PG}(d, q)$ in general. Run the same construction on lines against planes in $\text{PG}(3, q)$: two distinct lines either meet or are skew, so the planes containing both number 1 or 0, and the off-diagonal of $BB^\top$ takes two values differing by one — 9 and 10 for $q = 2$, 32 and 33 for $q = 3$, 144 and 145 for $q = 5$ (check (C)). Had the types of an $\tilde{A}_d$ complex put intermediate dimensions against each other, the correction term would have been a sum over the classes of the Grassmann association scheme instead of a multiple of $T^\top T$, and the remainder would have carried several moduli rather than one. Lemma 1.1 is what rules this out.

3. VERIFICATION

The script `ad_unitarity.py` runs the four checks of Table 1 in about three seconds; its output is archived as `ad_unitarity_output.txt`.

key	check	mode
(H)	$BB^\top = B^\top B = q^{d-1}I + q^{d-1}(q-1)J$ in $\text{PG}(d, q)$, 14 pairs (d, q)	exact
(C)	lines against planes in $\text{PG}(3, q)$: the off-diagonal takes two values	exact
(A)	the identities of §2 on Y_{21}, Y_{24}, Y_{42} , with $d = 2$	exact
(S)	exactly $ E - n$ singular values of L_E equal $q^{(d-1)/2}$	numeric

TABLE 1. The verification battery, 4/4.

Check (H) is exhaustive: for $d = 2, 3, 4, 5$ and $q = 2, 3, 5, 7, 11$ with $\theta_d \leq 1300$ it forms the full point/hyperplane non-incidence matrix of $\text{PG}(d, q)$ and both of its Gram matrices, and finds the diagonal constantly q^d and the off-diagonal constantly $q^{d-1}(q-1)$ — 14 instances, including $\text{PG}(5, 3)$ on 364 points and $\text{PG}(4, 5)$ on 781. Check (A) builds the thick $\tilde{A}_2$ complexes of $[X]$ and verifies Theorem 2.1, Proposition 2.3 and Corollary 2.4(ii) on each; check (S) finds 126, 144 and 252 singular values equal to $\sqrt{2}$ out of 147, 168 and 294, which are exactly $|E| - n$.

4. STATUS, SCOPE, CORRECTIONS

Proved. Lemma 1.1, Theorem 2.1, Proposition 2.3 and Corollary 2.4 for every $\tilde{A}_d$ complex of order q , $d \geq 2$; Corollary 2.5 for every such complex admitting an invariant $W \subseteq \ker S \cap \ker T$.

Not proved, and this is the gap. The existence of that W . For $d = 2$ it is [XV, Thm. 2.1], obtained from the intertwining $L_E \Psi = \Psi C'$ with $\Psi = [S^\top \ T^\top \ M^\top]$ and the cubic Hecke pencil. Corollary 2.4 is what survives unconditionally: L_E is already $q^{(d-1)/2}$ times an isometry from $\ker S$ onto $\ker T$, and only the passage from that pair of spaces to a single invariant one is missing.

Remark 4.1 (The candidate intertwiner in rank d). A chamber of an $\tilde{A}_d$ complex has $d+1$ vertices, so an edge $e = (x \rightarrow y)$ lying in a chamber determines $d-1$ further vertices, one of each type $\text{type}(y) + 1, \dots, \text{type}(y) + d-1$. Let $M_i(v, e)$ be the number of chambers containing e whose vertex of type $\text{type}(y) + i$ is v , and put

$$\Psi_d := [S^\top \ T^\top \ M_1^\top \ \dots \ M_{d-1}^\top] \in \mathbb{Z}^{|E| \times (d+1)n},$$

so that $\Psi_2 = [S^\top \ T^\top \ M^\top]$ is the intertwiner of [XV, Thm. 2.1] and the block count $d+1$ matches the degree of the Hecke pencil. We record this as the natural candidate and do not claim it works: what must be shown is $L_E \Psi_d = \Psi_d C^{(d+1)}$ and $L_E^\top \Psi_d = \Psi_d C^{(d+1)'} for $(d+1) \times (d+1)$ block matrices, after which $W = (\text{im } \Psi_d)^\perp$ would satisfy the hypothesis of Corollary 2.5 and complete$

the argument. Both identities are entrywise *local* — each entry is a count inside the star of a single vertex — so they can in principle be settled in $\mathrm{PG}(d, q)$ without a global complex, as Theorem 2.1 was.

Added in proof. That has now been carried out, in Paper XX. The local identity is a single flag count: in $\mathrm{PG}(d, q)$ the number of complete flags F with $F_{i+1} = V$ and F_1 inside a hyperplane H takes exactly two values, according as $V \subseteq H$ or not, and their difference $K_i = f(d-i)f(i+1)q^i/[i+1]_q$ depends on neither V nor H ; with $M_0 = cT$ and $M_d = cS$ this gives $L_E M_i^\top = K_i(T^\top A_{i+1} - c_{i+1}^{-1} M_{i+1}^\top)$ and, by the self-duality of $\mathrm{PG}(d, q)$, the mirror identity for $L_E^\top$. So Ψ_d does work, Corollary 2.5 is unconditional for every $d \geq 2$, and at $d = 2$ the characteristic polynomial of $C^{(3)}$ is the cubic Hecke pencil $x^3 - A_1 x^2 + q A_2 x - q^3$.

Remark 4.2 (Solved, open, impossible). *Solved:* the counting half of [XV, Rem. 6.1(d)] — the exact decomposition of $L_E L_E^\top$ in rank d — with the exponent $(d-1)/2$ as anticipated. *Open:* (a) *closed in Paper XX* — the invariant subspace $W_d = (\mathrm{im} \Psi_d)^\perp$ exists for every d , so Corollary 2.5 is unconditional; what remains open there is the exact rank of Ψ_d , hence $\dim W_d$; (b) a thick $\tilde{A}_3$ complex small enough to compute with, and with it the degree-four block companion and a worked example; (c) the exact certificate of the vertex Ramanujan property, the other half of [XV, Rem. 6.1(d)], which this note does not touch. *Impossible:* nothing new. In particular the mechanism does *not* degenerate in higher rank, which was the outcome one might have expected from Remark 2.6.

Remark 4.3 (Corrections to the task specification). Three. (1) The specification proposed building the identity from “the typical counting formula for the intersection of k -dimensional with $(d-k)$ -dimensional subspaces of $\mathrm{PG}(d, q)$ ”. No such formula is needed and none would serve: the typing of an $\tilde{A}_d$ complex puts hyperplanes against points (Lemma 1.1), and the only geometric input is that two distinct hyperplanes meet in codimension two. With intermediate dimensions the count is not constant (Remark 2.6), so an identity of the stated shape would be false. (2) The specification asked for $W_d = \bigcap_{i=0}^d \ker S_i$. Only two operators appear in Theorem 2.1, and $\ker S \cap \ker T$ already carries $RR^\top = q^{d-1}I$; whether a smaller intersection is needed is a question about *invariance*, not about the constant. (3) The predicted $R_d R_d^\top = q^{d-1}I$ and the circle $|u| = q^{-(d-1)/2}$ are correct, and are Corollary 2.5. The third task — the explicit $\tilde{A}_3$ block companion and a verified example — is not done: no thick $\tilde{A}_3$ complex is at hand. The rank-two test complexes came from an exhaustive search for triangle presentations over $\mathrm{PG}(2, 2)$ [X, Constr. 1.4]; there is no comparably small search in rank three, and the known constructions of finite $\tilde{A}_d$ quotients go through arithmetic lattices in PGL_{d+1} over a local field. That is a separate problem and we do not pretend to have solved it.

PROVENANCE

Unrefereed preprint. Theorem 2.1 and everything derived from it are proved in full; the computations verify the geometric input exhaustively in $\mathrm{PG}(d, q)$ for $d \leq 5$ and confirm that the rank- d statement specialises at $d = 2$ to the published identity on three complexes constructed independently. The one-line proof was found by asking which pair of types the $\mathbb{Z}/(d+1)$ grading actually selects, and the negative half of Remark 2.6 was computed before the positive half was proved, to check that the question had teeth.

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